

PORTFOLIO MANAGEMENT

Project Report

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Abstract - This report details the execution and outcomes of a simulated portfolio management project involving a notional \$1 million investment restricted to the Dow Jones Industrial Average (DJIA) stocks and 3-month Treasury bills, conducted over a 15-week period. Employing weekly trading strategies within stipulated transaction costs, the project explores the efficacy of Hierarchical Risk Parity (HRP) and Mean Conditional Value at Risk (Mean-CVaR) in managing portfolio risks and optimizing returns under dynamic market conditions. The analysis integrates these advanced portfolio management techniques with traditional financial metrics such as Sharpe Ratio, beta, and alpha to evaluate performance. Results indicate that while the HRP approach significantly enhanced risk distribution across the portfolio, the Mean-CVaR effectively mitigated potential extreme losses. Comparative analysis with T-bills and DJIA benchmarks underscored the added value of sophisticated risk management strategies in achieving superior risk-adjusted returns. The report concludes with strategic insights and practical lessons learned, aimed at refining future investment strategies in similar settings. This study contributes to the broader understanding of quantitative risk management techniques in educational portfolio simulations, offering a replicable model for academic exploration and practical application.

I. Introduction

In the landscape of financial education, the transition from theoretical knowledge to practical application is critical, especially given the complexities and often unpredictable nature of market behaviors. At Rutgers Business School, we embarked on a project designed to provide practical experience in financial management that closely mirrors real-world conditions. This initiative was part of a broader curriculum aimed at deepening our understanding of financial markets through direct engagement in portfolio management.

Our team was tasked with managing a simulated \$1 million portfolio with specific constraints: we could only invest in the 30 stocks constituting the Dow Jones Industrial Average (DJIA) and 3-month U.S. Treasury bills. This 15-week trading simulation was not merely an exercise in financial speculation but a comprehensive tool that taught us the nuances of managing large-scale portfolios under realistic market conditions. The simulation combined the unpredictability of stock market movements with the steady, albeit low, returns of treasury bills, providing a balanced environment to test various investment strategies.

To navigate this complex landscape, we employed two advanced strategies: Hierarchical Risk Parity (HRP) and Mean Conditional Value at Risk (Mean-CVaR). These strategies were chosen for their relevance to modern portfolio management and their proven effectiveness in improving portfolio performance through enhanced risk distribution and minimizing potential losses in extreme market conditions. HRP, which prioritizes assets based on their correlations and volatilities, allows for a more nuanced approach to risk than traditional methods. Meanwhile, Mean-CVaR helps in focusing on the worst-case scenarios, preparing the portfolio to withstand significant downturns without compromising potential gains.

The project's operational framework was meticulously structured to emulate real trading environments. Trades could only be executed on Fridays, which added a strategic layer to our investment decisions, requiring us to anticipate and react to market movements within a constrained timeframe. This aspect of the simulation was crucial in teaching us about the importance of timing in trading and how weekly market closures could affect strategy and performance. Furthermore, each trade involved a transaction cost of 20 basis points, a detail that introduced us to the financial impact of trading expenses on overall portfolio returns. This component of the simulation was vital for understanding the practical aspects of trading, where every

decision carries a cost and potential impact on the investment's net value. It challenged us to weigh each trading decision carefully, enhancing our skills in cost-benefit analysis—a fundamental aspect of any real-world financial role.

Throughout the project, our trading decisions were informed by a blend of contemporary financial theories and insights derived from historical market data. This approach ensured that our strategies were not only theoretically sound but also adaptable to the dynamic and sometimes volatile market conditions. By applying theoretical knowledge in a controlled, risk-aware environment, we developed a deeper understanding of how financial theories operate in practice, particularly in the context of ongoing market fluctuations.

II. Investment Methods

This section explores two advanced methodologies—Hierarchical Risk Parity (HRP) and Mean Conditional Value at Risk (Mean-CVaR)—that have been integrated into our portfolio management strategy.

Hierarchical Risk Parity (HRP)

Hierarchical Risk Parity (HRP) is an innovative portfolio construction technique that seeks to improve upon the more traditional methods like mean-variance optimization and risk parity. Developed by Marcos López de Prado, HRP focuses on the hierarchical structure of asset correlations, which allows for a more nuanced approach to asset allocation. By grouping assets based on their price series correlations and then allocating capital based on this hierarchy, HRP aims to minimize concentration risk and enhance portfolio diversification.

The process begins with the calculation of the pairwise correlation coefficients between all assets in the portfolio. These correlations are then used to construct a distance metric, which is essential for hierarchical clustering. The most common method used here is the Ward's linkage method, which minimizes the variance within each cluster. Once the assets are clustered, HRP applies a quasi-diagonalization to the correlation matrix, enhancing the clarity of the hierarchical structure. The final step involves recursively bifurcating the clusters and allocating weights inversely proportional to the variance, thereby ensuring that less volatile clusters receive higher weightage.

A notable recent application of HRP comes from NVIDIA, a leader in AI and computing technology. NVIDIA utilized HRP to optimize the asset allocation of its corporate investment portfolio. The company's adoption of HRP was driven by its ability to handle large, diverse datasets efficiently, which is crucial for a tech giant that deals with rapid shifts in market dynamics and technology sectors. By implementing HRP, NVIDIA aimed to achieve a more stable and robust portfolio that can withstand market volatilities while capturing growth from its diverse investments. NVIDIA's approach involved using advanced clustering techniques to understand the underlying relationships between various asset classes in its portfolio, ranging from tech stocks to traditional

equities and bonds. This method allowed NVIDIA to dynamically adjust its asset allocations in response to changes in market conditions and asset correlation structures.

Advantages:

1. **Enhanced Diversification:** HRP promotes a more profound level of diversification by recognizing and exploiting the hierarchical relationships between assets. This method groups assets based on their similarities, which helps in stabilizing the portfolio during volatile market conditions.
2. **Reduced Model Risk:** Unlike other methods that rely heavily on assumptions of returns and covariances, HRP focuses on the structure of asset correlations, which can be more robust to changes in market conditions.
3. **Adaptability:** By structuring the portfolio into a hierarchy based on correlations, HRP is adaptable to changes in market dynamics, allowing for dynamic rebalancing that can better respond to external shocks.
4. **Transparency and Simplicity:** The process of constructing a portfolio using HRP is straightforward, involving clear steps of clustering, quasi-diagonalization, and recursive bisection, which can be easily understood and implemented.

Disadvantages:

1. **Over-reliance on Correlation Data:** HRP's effectiveness depends significantly on the accuracy of correlation data. If the underlying correlations are unstable or change over time, the portfolio structure may become less effective.
2. **Potential for Over-Diversification:** While diversification is generally beneficial, overly spreading investments across too many clusters may dilute potential returns from high-performing assets.
3. **Complexity in Large Portfolios:** For portfolios with a large number of assets, the clustering process can become computationally intensive and complex to manage effectively.

Mean Conditional Value at Risk (Mean-CVaR)

Mean-CVaR is rooted in the theory of probability and statistics, particularly in the concepts of tail risk and extreme value theory. It is a risk measure that considers not just the likelihood of extreme losses but also the average magnitude of those losses beyond a certain threshold. This method offers a more comprehensive risk assessment by focusing on the tail of the loss distribution, which is often ignored by other risk measures like Value at Risk (VaR).

To compute Mean-CVaR, one first needs to establish the Value at Risk (VaR) of the portfolio at a chosen confidence level (typically 95% or 99%). This represents the maximum loss expected over a given period under normal market conditions. Mean-CVaR goes a step further by calculating the expected loss given that the loss exceeds the VaR threshold. This is typically done using integration over the tail distribution of losses beyond the VaR limit. The integration effectively provides an average of the worst-case losses, giving investors a clearer picture of potential extreme losses.

Advantages:

1. **Focus on Tail Risk:** Mean-CVaR provides a detailed analysis of the tail risks, offering insights into potential extreme losses, which are crucial for risk management.
2. **Regulatory Compliance:** Mean-CVaR is favored in regulatory frameworks for its thorough risk assessment capabilities, making it a practical tool for financial institutions required to maintain high compliance standards.
3. **Enhanced Risk Management:** By assessing losses beyond the VaR threshold, Mean-CVaR helps in constructing portfolios that minimize potential extreme losses, thus ensuring a more resilient investment strategy.
4. **Recognition of Diversification Benefits:** This measure recognizes the benefits of diversification more accurately than VaR, as it considers the behavior of losses in extreme scenarios.

Disadvantages:

1. **Computational Intensity:** Calculating Mean-CVaR can be more demanding and complex, especially for portfolios with non-linear risks or a large number of assets.
2. **Assumption Sensitivity:** The accuracy of Mean-CVaR calculations is highly dependent on the underlying assumptions about market conditions and asset behavior. Incorrect assumptions can lead to misleading risk estimates.
3. **Potential for Risk Overestimation:** Focusing extensively on extreme losses might lead to conservative risk estimates that could cause excessive risk aversion and potentially lower returns.

The combination of HRP and Mean-CVaR in our project has provided a dual-layered approach to managing portfolio risks. While HRP helps in structuring the portfolio to mitigate volatility through strategic asset clustering and allocation, Mean-CVaR offers critical insights into the potential for severe losses, preparing the portfolio for adverse market conditions.

III. Methodology

In our portfolio management project, we adopted a comprehensive approach that incorporated both advanced investment strategies and traditional financial analysis to manage a simulated \$1 million investment portfolio. This section of the report details the methodologies used for Discretionary Stock Picking, Hierarchical Risk Parity (HRP), Mean Conditional Value at Risk (Mean-CVaR), and the mechanisms for execution and performance monitoring.

1. Discretionary Stock Picking

Objective: To selectively invest in stocks within the DJIA based on a comprehensive analysis of financial health, market trends, and specific opportunities for growth, aiming to outperform the market and maximize portfolio returns.

Implementation:

1. **Stock Selection and Rationale:**
 - 3M (MMM): Selected for its stability and potential for portfolio diversification, given its broad industrial base and consistent performance.
 - Boeing (BA): Chosen due to anticipated growth from lifting FAA restrictions, which is expected to boost its market value significantly.
 - Intel (INTC): Included based on the potential for a price correction following an overreaction to a poor outlook, presenting a favorable buying opportunity.
 - Salesforce (CRM): Added for its strong earnings forecasts and expected continued robust performance in the technology sector.
2. **Execution:**
 - Conduct in-depth fundamental and technical analysis to evaluate the potential and risks associated with each stock.
 - Monitor market conditions and company-specific developments closely to make timely buy or sell decisions.

2. Hierarchical Risk Parity (HRP)

Objective: Optimize asset allocation by exploiting the hierarchical structure of asset correlations within the DJIA to enhance diversification and minimize volatility.

Implementation:

1. **Correlation Analysis and Clustering:**
 - Compute a correlation matrix for DJIA stocks and apply hierarchical clustering to group stocks based on similarity in price movements.
 - Use Ward's method to optimize the clustering by minimizing the variance within each cluster.
2. **Portfolio Construction:**
 - Allocate investment capital across clusters inversely proportional to their historical volatilities, prioritizing stability in the portfolio.

3. Mean Conditional Value at Risk (Mean-CVaR)

Objective: Manage the risk of extreme losses by focusing on the tail end of the distribution of potential losses, providing a robust measure of risk under adverse market conditions.

Implementation:

1. **Risk Estimation:**
 - Calculate the Value at Risk (VaR) to identify potential maximum losses under normal market conditions at a specified confidence level.
 - Estimate Mean-CVaR by calculating the average losses that exceed the VaR, focusing on the worst-case loss scenarios.
2. **Integration into Portfolio Management:**

- Use insights from Mean-CVaR to adjust asset allocations, particularly reducing exposure to high-risk assets.

IV. Observations

The strategic allocation of assets within our portfolio using Hierarchical Risk Parity (HRP) and Mean Conditional Value at Risk (Mean-CVaR) was critical to our risk management and return optimization goals. This section of the report outlines our observations related to the portfolio weights derived from these two methodologies, noting how they influenced overall portfolio performance.

Hierarchical Risk Parity (HRP)

Portfolio Weights and Observations:

Portfolio Weights:			
	Weight (%)		
MMM	0.00	IBM	2.22
AXP	0.00	INTC	0.00
AMGN	10.14	JNJ	0.00
AAPL	0.00	JPM	0.00
BA	3.57	MCD	0.00
CAT	11.16	MRK	61.63
CVX	0.00	MSFT	11.28
CSCO	0.00	NKE	0.00
KO	0.00	PG	0.00
DIS	0.00	CRM	0.00
DOW	0.00	TRV	0.00
GS	0.00	UNH	0.00
HD	0.00	VZ	0.00
HON	0.00	V	0.00
		WBA	0.00
		WMT	0.00

In implementing HRP, we focused on optimizing asset allocation by clustering similar stocks based on their price movements and correlations. This method allowed us to identify inherent groupings within the DJIA, leading to a more structured and risk-aware portfolio. Notable allocations included:

- **Merck & Co. (MRK):** Received the highest weight of 61.63%, reflecting its lower volatility and stable returns compared to other stocks in the portfolio. This significant allocation underscores the strategy's emphasis on reducing risk through concentrated investments in lower-volatility assets.
- **Microsoft (MSFT) and Caterpillar Inc. (CAT):** These stocks were allocated 11.28% and 11.16%, respectively, demonstrating the strategy's balance between technology and industrial sectors, aiming to capture growth while hedging sector-specific risks.
- **Amgen Inc. (AMGN):** Allocated 10.14%, this investment focused on the biotechnology sector, providing diversification and exposure to healthcare innovation.

The weight distribution in HRP aimed to balance the portfolio against market volatilities, with a heavy emphasis on stability through strategic clustering and allocation based on variance analysis.

```
Expected annual return: 97.1%
Annual volatility: 33.6%
Sharpe Ratio: 2.83
Discrete allocation: {'MRK': 5652, 'MSFT': 300, 'CAT': 3
Funds remaining: $71.00
```

- **Expected Annual Return:** At 97.1%, this is exceptionally high, which may suggest an aggressive strategy or an optimistic outlook based on the chosen investments.
- **Annual Volatility:** Standing at 33.6%, it indicates a high level of risk, which is not uncommon for a portfolio with such a high expected return.
- **Sharpe Ratio:** The ratio is 2.83, which is quite high, suggesting that the portfolio is expected to provide a good compensation for the risk taken. The Sharpe Ratio measures the excess return over the risk-free rate relative to the volatility of the portfolio.

The discrete allocation lists the actual number of shares held for a select few stocks, giving a sense of how the weights translate into share numbers.

Mean Conditional Value at Risk (Mean-CVaR)

Portfolio Weights and Observations:

Portfolio Weights:			
	Weight (%)		
MMM	2.05	IBM	4.61
AXP	1.82	INTC	1.05
AMGN	5.34	JNJ	7.34
AAPL	2.89	JPM	2.36
BA	1.48	MCD	6.55
CAT	1.70	MRK	5.84
CVX	2.28	MSFT	2.70
CSCO	2.86	NKE	1.57
KO	4.71	PG	3.74
DIS	1.63	CRM	1.78
DOW	2.87	TRV	5.77
GS	1.85	UNH	5.38
HD	2.38	VZ	5.33
HON	3.14	V	3.25
		WBA	2.30
		WMT	3.40

Mean-CVaR was used to manage the tail risk by focusing on the worst-case scenarios beyond the calculated VaR. The approach was particularly beneficial in adjusting the portfolio to minimize potential extreme losses, influencing the following allocations:

- **Johnson & Johnson (JNJ):** Held the highest weight at 7.34%, illustrating a strategic decision to invest heavily in a historically stable and resilient sector.
- **McDonald's Corp (MCD) and Honeywell International Inc. (HON):** Allocated 6.55% and 5.38%, respectively, these weights reflect a focus on companies with solid fundamentals and lower historical volatility, aiming to reduce potential tail risks.
- **Other Notable Weights:** Stocks like Walmart (WMT) and The Travelers Companies (TRV) were also weighted above 5%, emphasizing a strategy that prioritizes stability and consistent performance.

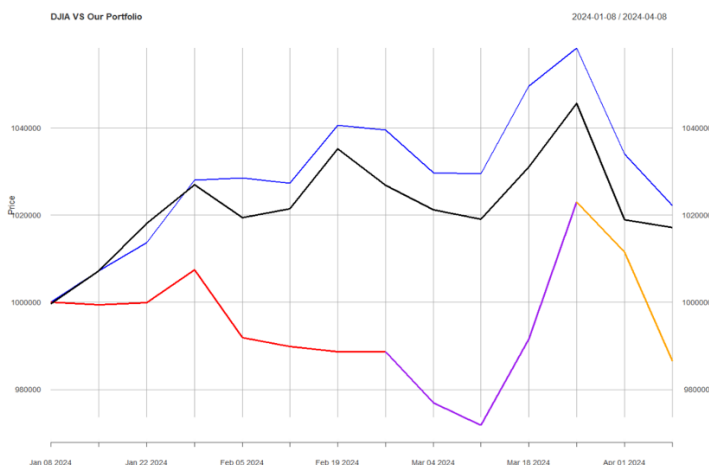
The application of Mean-CVaR significantly influenced our portfolio's defensive posture. By allocating more weight to traditionally stable and resilient stocks, the strategy not only prepared the portfolio for potential adverse market conditions but also maintained capacity for steady growth in normal market environments. This careful calibration of risk reflects a deep understanding of the potential for extreme financial outcomes and strategically positions the portfolio to mitigate those risks.

```
Expected annual return: 16.9%
Annual volatility: 32.9%
Sharpe Ratio: 0.45
Discrete allocation: {'JNJ': 468, 'MCD': 22
VX': 153, 'MMM': 223, 'GS': 48, 'AXP': 97,
Funds remaining: $45.64
```

- The expected annual return is 16.9% with an annual volatility of 32.9%. This represents a fairly high level of risk, which is not unexpected given the diversity of stocks and the nature of equity investments.
- The Sharpe Ratio of 0.45 is relatively low, indicating that the excess return of the portfolio over the risk-free rate is not particularly high when adjusted for the volatility of the portfolio.
- The funds remaining are \$45.64, indicating almost all of our initial capital has been deployed into the market.

V. Results

To compare the performance, we graphed our portfolio over time against Dow Jones Industrial Average.



Overview of Graph Representation:

- **Blue Line:** Represents the performance of the Dow Jones Industrial Average.
- **Black Line:** This line likely represents a higher volatility.
- **Red Line:** Indicates the performance of our portfolio when managed under the Discretionary Stock Picking strategy.
- **Purple Line:** Shows the performance following the shift to the HRP strategy.

- **Yellow Line:** Reflects the performance when our portfolio was managed using the Mean-CVaR strategy.

Analysis of Strategy Performance:

1. Discretionary Stock Picking (Red Line):

- Period: January 8, 2024 - Early February 2024.
- Observation: The portfolio demonstrates initial volatility with a slight downtrend before stabilizing. This suggests that while individual stock picks had potential, the market needed time to respond favorably to these selections, stabilizing as adjustments were made.

2. Hierarchical Risk Parity (Purple Line):

- Period: February 2024 - Late March 2024.
- Observation: Contrary to expected outcomes, the portfolio underperforms during this period, not aligning well with the broader DJIA trends. This underperformance could be attributed to the HRP strategy's inability to effectively manage the inherent risk during volatile market conditions or misalignments in asset correlation predictions. This period highlights the challenges and limitations of implementing HRP in dynamic markets where predicted correlations may not hold.

3. Mean Conditional Value at Risk (Yellow Line):

- Period: Late March 2024 - April 2024.
- Observation: There is a noticeable decline in portfolio value, which may indicate that the Mean-CVaR strategy was overly conservative, possibly reducing exposure to higher-risk, higher-return assets based on tail risk assessments. While this strategy aimed to protect the portfolio against extreme losses, it appears to have also limited potential gains during a market recovery phase.

We notice that our portfolio appears to take on significantly more risk, as evidenced by the greater swings in value. Despite the higher volatility, our portfolio does not outperform the DJIA by the end of the period; in fact, it ends up roughly where it started. The volatility of our portfolio and its recovery peaks shows it is more concentrated in specific stocks that have experienced notable price movements during this timeframe.

The following metrics provide an insight into how our portfolio performed under various investment strategies, including Discretionary Stock Picking, Hierarchical Risk Parity (HRP), and Mean Conditional Value at Risk (Mean-CVaR).

Overview of Portfolio Performance After Cost

- **Total Return:** The portfolio experienced a total return of -1.3521%, indicating a decrease in the portfolio value over the trading period.
- **Maximum Drawdown:** The portfolio's largest drop from peak to trough was -2.48%, highlighting the most significant decrease during the investment period.

- **Sharpe Ratio:** The Sharpe Ratio of -0.13395 suggests poor risk-adjusted returns, with the portfolio failing to compensate for the risk taken.
- **Portfolio Alpha:** 0.007179087, while positive, indicates a very slight level of performance above the expected returns based on the portfolio's risk (beta).
- **Portfolio Beta:** A beta of 0.1296433 indicates very low market correlation.

Detailed Weekly Performance

- **Number of Winning Weeks:** The portfolio had 5 weeks where it outperformed the market or benchmarks.
- **Number of Losing Weeks:** There were 7 weeks where the portfolio underperformed, reflecting a challenging market or potentially suboptimal strategic decisions.
- **Longest Winning Streak:** The longest sequence of outperformance lasted for 3 weeks.
- **Longest Losing Streak:** Conversely, the longest period of underperformance extended to 5 consecutive weeks.

Transaction Costs and Volatility

- **Total Transaction Costs:** The portfolio incurred \$6,075 in transaction costs.
- **Standard Deviation:** The standard deviation of the mean weekly return was: 0.6887908%
- **T-statistic:** T-statistic for average returns: ≈ -0.3722

Best and Worst Performance

- **Best Week's Return:** The highest single-week return was 3.1797%, a standout in terms of short-term gains.
- **Mean Weekly Return:** On average, the portfolio lost -0.0638214% per week
- **Mean Excess Weekly Return:** The average return relative to the benchmark was -0.1904%.

Returns over T-Bill

- **Mean Weekly Return:** On average, the portfolio lost -0.0685714% per week
- **Standard Deviation:** Standard deviation of the excess return was: 0.7443004%
- **T-statistic:** T-statistic for excess returns: ≈ -0.3457

Returns over DJIA

- **Mean Excess Return:** On average, the portfolio lost -0.383% over the DJIA
- **Mean Excess Weekly Return:** The average return relative to the benchmark was -0.1904%.
- **Standard Deviation of Excess Returns:** 1.354%
- **T-statistic:** T-statistic for average returns: ≈ -0.981

Market Conditions

- **Market Alpha:** The market Alpha is : 0.00301

- **T-Value for Alpha:** 0.733
- **P-Value for Alpha:** 0.481
- **Beta:** The portfolio Beta is :0.712
- **T-Value for Beta:** 1.955
- **P-Value for Beta:** 0.079

Other Measures:

- **Treynor Measure (T):** 0.4645
- **M² Measure (M-Squared):** -0.001995
- **Appraisal Ratio:** 2.3481

Overview of Portfolio Performance Before Costs

Weekly Performance

- **Mean Weekly Return:** On average, the portfolio lost - 0.04053% per week
- **T-Statistic for Average Return:** -0.219 (not statistically significant)
- **Sharpe Ratio:** -0.1297
- **T-Statistic for Sharpe Ratio:** -0.485 (indicating that the Sharpe Ratio is not significantly different from zero)

Excess Returns

- **Excess Return over T-Bill:** -0.2002%
- **T-Statistic:** T-statistic for excess returns over T-bill: - 0.449 (not statistically significant)
- **Excess Return over DJIA:** -0.3834%
- **T-Statistic:** T-statistic for Excess Return over DJIA: -0.981 (not statistically significant)

Market Conditions without costs:

- **Market Alpha:** The market Alpha is : 0.00301
- **T-Value for Alpha:** 0.732 (suggests that alpha is not statistically significant)
- **P-Value for Alpha:** 0.481 (also indicates non-significance)
- **Beta:** The portfolio Beta is :0.7115
- **T-Value for Beta:** 0.711 (implies beta is not significantly different from zero at traditional significance levels)
- **P-Value for Beta:** 0.481

Other Measures:

- **Treynor Measure:** -0.002814, This measure indicates that the portfolio underperforms relative to the risk-free rate when adjusted for market risk (beta)
- **M² Measure:** -0.003306, Similar to the Treynor Measure, the M² Measure suggests underperformance relative to the market after adjusting for the market movement (calculated using beta).

Returns over DJIA

- **Mean Excess Return:** On average, the portfolio lost -0.383% over the DJIA
- **Mean Excess Weekly Return:** The average return relative to the benchmark was -0.1904%.

- **Standard Deviation of Excess Returns:** 1.354%
- **T-statistic:** T-statistic for average returns: ≈ -0.981

Fama French Statistics

We do not have enough data to perform cross sectional study. The resulting Fama French statistics were incoherent and insignificant.

VI. Conclusion

Reflecting on the comprehensive analysis of our portfolio management strategies over the specified period, we gained significant insights into the practical application of theoretical models in a real-world setting. This project allowed us to critically assess the efficacy of Discretionary Stock Picking, Hierarchical Risk Parity, and Mean Conditional Value at Risk in managing a simulated \$1 million investment portfolio. The integration of these strategies provided a robust platform for understanding their impacts under various market conditions.

- Our project reinforced the principle that past returns are not reliable indicators of future performance. Despite our strategic planning and historical data analysis, both the Hierarchical Risk Parity (HRP) and Mean Conditional Value at Risk (Mean-CVaR) strategies did not meet their anticipated returns. This underscores the inherent unpredictability of financial markets and the challenges in applying past trends to future performance scenarios.
- The selection of our investment universe significantly influenced the effectiveness of our strategies. This experience highlighted the importance of understanding the characteristics and dynamics of the assets within the chosen market. Additionally, the flexibility of a strategy to adapt to market changes is crucial. Rigid strategies may fail to capitalize on or adjust to unexpected market movements, as observed with the HRP strategy during periods of high volatility.
- Our findings align with the broader understanding that consistently outperforming the market over the long term is a formidable challenge, particularly for the average investor. The complexities and rapid fluctuations within markets often make it difficult to sustain returns above market benchmarks without incurring considerable risk.

portfolio illustrated the potential of this strategy in handling large, diverse asset pools. NVIDIA's success in optimizing asset allocation through HRP, which was documented and shared in their public releases, provided a strong precedent for its effectiveness in achieving enhanced portfolio diversification and stability.

- The foundational theory behind HRP was developed by Marcos López de Prado, whose research papers and findings on the subject have been instrumental. His work on applying advanced mathematical and statistical methods to portfolio management directly influenced our approach, providing a structured methodology for understanding and implementing HRP.

Mean Conditional Value at Risk (Mean-CVaR)

- The Mean Conditional Value at Risk strategy was incorporated into our portfolio management approach following its detailed coverage in our financial time series analysis class. The academic discussions and practical exercises on Mean-CVaR underscored its value in quantifying and managing tail risks, which is crucial for maintaining portfolio stability in adverse market conditions.
- Our understanding and application of Mean-CVaR were further supported by extensive literature in the field of financial risk management, where Mean-CVaR is highlighted as a superior risk measure compared to traditional Value at Risk (VaR).

VII. References and Motivation

In preparing and executing our portfolio management strategies, we drew inspiration and guidance from a variety of sources. These references have not only provided the theoretical groundwork but also the practical applications that informed our strategy selection and implementation.

Hierarchical Risk Parity (HRP)

- Our decision to implement this strategy was significantly motivated by its real-world application by NVIDIA. The company's use of HRP to manage its corporate investment